전산화학 (2017년 1학기) **1차 시험 해답**

(총 5 문제, 65점)

1. **(총 15점, 각 5점)**

(a) $z = 4e^{3i} + 5e^{2i}$

$z = 4e^{3i}+5e^{2i} = 4[\cos(3)+i\sin(3)]+5[\cos(2)+i\sin(2)] = [4\cos(3)+5\cos(2)+i[4\sin(3)+5\sin(2)]$

$\therefore \mathrm{Re}(z) = 4\cos(3) + 5\cos(2), \qquad \mathrm{Im}(z) = 4\sin(3) + 5\sin(2)$ 이다.

(b) $z = (a + ib)^2 - 4e^{iab}$ (단, a, b, c, d 는 상수)

$z = (a+ib)^2-4e^{iab} = a^2-b^2+i(2ab)-4\cos(ab)-4[\cos(ab)+i\sin(ab)] = [a^2-b^2-4\cos(ab)]+i[2ab-4\sin(ab)]$

$\therefore \mathrm{Re}(z) = a^2 - b^2 - 4\cos(ab), \qquad \mathrm{Im}(z) = 2ab - 4\sin(ab)$ 이다.

(c) $-1+\sqrt{3}i = 2\Big(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\Big) = 2\big[\cos(2n\pi+2\pi/3)+i\sin(2n\pi+2\pi/3)\big] = 2e^{i(2n\pi+2\pi/3)}$
(단, n 은 정수) 이므로

$$\begin{aligned} z = (-1+\sqrt{3}i)^{1/2} &= \left(2e^{i(2n\pi+2\pi/3}\right)^{1/2} \\ &= \sqrt{2}e^{i(n\pi+\pi/3)} \\ &= \sqrt{2}\left[\cos\left(n\pi+\frac{\pi}{3}\right)+i\sin\left(n\pi+\frac{\pi}{3}\right)\right] \end{aligned} \tag{1}$$

i. $n = 0$ 인 경우, (1) 식은

$$z = \sqrt{2}\left[\cos\left(\frac{\pi}{3}\right)+i\sin\left(\frac{\pi}{3}\right)\right] = \frac{\sqrt{2}}{2}+\frac{\sqrt{6}}{2}i$$

$\therefore \mathrm{Re}(z) = \frac{\sqrt{2}}{2}, \qquad \mathrm{Im}(z) = \frac{\sqrt{6}}{2}$ 이다.

ii. $n = 1$ 인 경우, (1) 식은

$$z = \sqrt{2}\left[\cos\left(\frac{4\pi}{3}\right)+i\sin\left(4\frac{\pi}{3}\right)\right] = -\frac{\sqrt{2}}{2}-\frac{\sqrt{6}}{2}i$$

$\therefore \mathrm{Re}(z) = -\frac{\sqrt{2}}{2}, \qquad \mathrm{Im}(z) = -\frac{\sqrt{6}}{2}$ 이다.

2. **(총 10점, 각 5점)**

(a) $z = 4e^{3i} + 6i$

$$z = 4e^{3i} + 6i = 4(\cos 3 + i\sin 3) + 6i = 4\cos 3 + i(\sin 3 + 6)$$

$\therefore z^* = \cos 3 - i(\sin 3 + 6)$

(b) $z = (2 + i)^2 - (7i)^2$

$$z = (2+i)^2 - (7i)^2 = 4 + 4i - 1 + 49 = 52 + 4i$$

$\therefore z^* = 54 - 4i$

3. **(총 15점, 각 5점)**

(a) $(r, \theta) = (2, 5\pi/3)$

$$x = r\cos\theta = 2 \times \cos\left(\frac{5\pi}{3}\right) = 1$$
$$y = r\sin\theta = 2 \times \sin\left(\frac{5\pi}{3}\right) = -\sqrt{3}$$

(b) $(x, y, z) = (\sqrt{3}, -1, 0)$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{\sqrt{3}^2 + (-1)^2 + 0^2} = 2$$
$$\theta = \arccos\left(\frac{z}{r}\right) = \arccos\left(\frac{0}{2}\right) = \arccos(0) = \frac{\pi}{2}$$
$$\phi = \arctan\left(\frac{y}{x}\right) = \arctan\frac{-1}{\sqrt{3}} = \frac{11\pi}{6}$$

(c) $(r, \theta, \phi) = (2, 2\pi/3, 5\pi/4)$

$$x = r\sin\theta\cos\phi = 2 \times \sin\left(\frac{2\pi}{3}\right) \times \cos\left(\frac{5\pi}{4}\right) = 2 \times \frac{\sqrt{3}}{2} \times -\frac{\sqrt{2}}{2} = -\frac{\sqrt{6}}{2}$$
$$y = r\sin\theta\sin\phi = 2 \times \sin\left(\frac{2\pi}{3}\right) \times \sin\left(\frac{5\pi}{4}\right) = 2 \times \frac{\sqrt{3}}{2} \times -\frac{\sqrt{2}}{2} = -\frac{\sqrt{6}}{2}$$
$$z = r\cos\theta = 2 \times \cos\left(\frac{2\pi}{3}\right) = 2 \times -\frac{1}{2} = -1$$

4. **(총 15점, 각 5점)**

(a) 문제의 (1) 식을 다시 쓰면

$$\begin{aligned}\psi_{311}(r, \theta, \phi) &= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\left(6 - \frac{r}{a_0}\right)e^{-r/(3a_0)}\sin\theta e^{i\phi} \\ &= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\left(6 - \frac{r}{a_0}\right)e^{-r/(3a_0)}\sin\theta\big(\cos\phi + i\sin\phi\big)\end{aligned} \tag{2}$$

그러므로

$$\begin{aligned}\mathrm{Re}(\psi_{311}) &= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\left(6 - \frac{r}{a_0}\right)e^{-r/(3a_0)}\sin\theta\cos\phi, \\ \mathrm{Im}(\psi_{311}) &= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\left(6 - \frac{r}{a_0}\right)e^{-r/(3a_0)}\sin\theta\sin\phi,\end{aligned}$$

(b)

$$|\psi_{311}(r, \theta, \phi)|^2 = \psi^*_{322}\psi_{322} = \frac{1}{81^2\pi}\left(\frac{1}{a_0}\right)^3\left(\frac{r}{a_0}\right)^2\left(6 - \frac{r}{a_0}\right)^2 e^{-2r/(3a_0)}\sin^2\theta$$

(c) (a)의 (3) 식으로부터

$$\begin{aligned}\psi_{311}(r, \theta, \phi) &= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\left(6 - \frac{r}{a_0}\right)e^{-r/(3a_0)}\sin\theta e^{i\phi} \\ &= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{1}{a_0}\left(6 - \frac{r}{a_0}\right)e^{-r/(3a_0)}r\sin\theta e^{i\phi} \\ &= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{5/2}\left(6 - \frac{r}{a_0}\right)e^{-r/(3a_0)}r\sin\theta\big(\cos\phi + i\sin\phi\big) \\ &= \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{5/2}\left(6 - \frac{r}{a_0}\right)e^{-r/(3a_0)}\big(r\sin\theta\cos\phi + ir\sin\theta\sin\phi\big)\end{aligned} \tag{3}$$

3차원 직교 좌표계와 구면 극 좌표계의 관계식으로부터

$$\begin{aligned} x &= r\sin\theta\cos\phi \\ y &= r\sin\theta\sin\phi \\ r &= \sqrt{x^2+y^2+z^2} \end{aligned}$$

이므로, 관계식을 위의 (3) 식에 대입하면

$$\psi_{311}(x,y,z) = \frac{1}{81\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{5/2}\left(6-\frac{\sqrt{x^2+y^2+z^2}}{a_0}\right)e^{-\sqrt{x^2+y^2+z^2}/(3a_0)}\big(x+iy\big) \tag{4}$$

5. **(총 10점)** 주어진 문제의 식의 양 변에 자연로그를 취하면,

$$-\sigma bc = -\epsilon bc\ln 10 \tag{5}$$

그러므로 $\sigma = \epsilon\ln 10$.